

Electricity and Magnetism

MP

Teach You Physics Hand In Hand

March 2026

1 SOLUTION to Problem in Circuit Post

- Thanks to buti for proposing the question.
- Level of difficulty of the problem: **2/10**
- Assumed "ideal" transformer, so $M = L$.

Notice that $\frac{d}{dt}\tilde{I}(t) = \frac{d}{dt}I_0e^{i\omega t} = I_0\frac{d}{dt}e^{i\omega t} = (i\omega)I_0e^{i\omega t} = i\omega\tilde{I}(t)$

Using complex voltage, Kirchoff's Rules gives: [8 points]

$$\begin{cases} \tilde{V}(t) - i\omega L\tilde{I}_p(t) - i\omega M\tilde{I}_s(t) = 0 \\ \tilde{I}_s(t)R + i\omega L\tilde{I}_s(t) + i\omega M\tilde{I}_p(t) = 0 \end{cases}$$

Solving simultaneous equations, we get the complex form of secondary current:

$$\Rightarrow \tilde{I}_s(t) = \frac{i\omega MV_0}{(i\omega L)(R + i\omega L) + \omega^2 M^2} e^{i\omega t}$$

And the real form (Answer of part (a)): [2 points]

$$\Rightarrow I_s(t) = \frac{MV_0}{\sqrt{\omega^2(M^2 - L^2)^2 + L^2 R^2}} \cos(\omega t) = \frac{V_0}{R} \cos(\omega t)$$

By root-mean-square method, average power (Answer of part (b)): [10 points]

$$\Rightarrow \langle P \rangle = I_{(s)r.m.s}^2 R = \frac{1}{2} \frac{RM^2 V_0^2}{\omega^2(M^2 - L^2)^2 + L^2 R^2} = \frac{V_0^2}{2R}$$

Of course you can use **time-averaging integral** to get the same result.

$$\langle P \rangle = \frac{1}{T} \int_0^T I_s^2 R dt = \frac{1}{T} \int_0^T R \left[\frac{MV_0}{\sqrt{\omega^2(M^2 - L^2)^2 + L^2 R^2}} \cos(\omega t) \right]^2 dt = \frac{V_0^2}{2R}$$

2 Maxwell's Equations

By experiment, we can derive the Maxwell's Equations in differential form:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad (1)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (3)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (4)$$

Theoretically, for any problem in EM, you can solve differential form of Maxwell's Equations (PDEs) to get an answer, but that will be too annoying. Therefore, we often use integral form:

$$\oiint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\varepsilon_0}, \quad (5)$$

$$\oiint \mathbf{B} \cdot d\mathbf{A} = 0, \quad (6)$$

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{\partial}{\partial t} \iint \mathbf{B} \cdot d\mathbf{A}, \quad (7)$$

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \iint \mathbf{E} \cdot d\mathbf{A}. \quad (8)$$

where:

- (1) and (5) = Gauss Law for E field.
- (2) and (6) = Gauss Law for B field.
- (3) and (7) = Faraday's Law.
- (4) and (8) = Ampere's Law.

Additional notes:

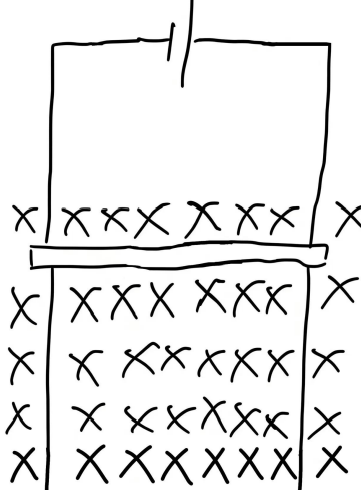
- There is another very useful formula for B fields called the Biot-Savart Law, which describes the magnetic field $\mathbf{B}$ generated by a steady current. It is a fundamental law of magnetostatics.

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} dV'$$

- $\mathbf{B}$ field for magnetic dipole: For an ideal magnetic dipole with dipole moment $\mathbf{m}$ located at the origin, the magnetic field at $\mathbf{r}$ is given by:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi r^3} [3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{m}) - \mathbf{m}]$$

3 Problem



A rod of mass m and length l is falling vertically in an infinitely long smooth U-shape wire frame with a resistive force $F = -\beta v$. It is known that during the falling process, the rod is always perpendicular to a uniform magnetic field $\vec{B}$. Assume that the total resistance R of the circuit is constant.

The following parameters are given: $y(0) = y_0$, $\dot{y}(0) = 0$, $V_{\text{battery}} = V_0$

- [10 points] Derive and solve for the equation of motion of the rod, i.e. $y(t)$.
- [10 points] Find the distance travelled by the rod at the instant when the rod attains $x\%$ of its terminal velocity.

Notes: You may consider the following substitutions to make your calculations prettier: $\alpha = \frac{\beta}{m} + \frac{B^2 l^2}{mR}$, $\gamma = g + \frac{BlV_0}{mR}$.

The solution will be uploaded in the next post and MP love you all :)